

Scope and instructions

SCOPE AND INSTRUCTIONS

Concepts needed: representation; explanation; equations; diagrams

This answer key treats a conceptual response as a chain of ideas rather than a single keyword. Full credit requires the physical object, the relationship, and the direction or limiting behavior. Diagrams should be judged by structure: axes and variables, arrows, phase, and labels matter more than artistic style.

1. Use the word bank only where requested; repeated terms are intentional.
2. For explanation prompts, name the governing principle before applying it.
3. For drawings, accept equivalent orientation if the relative geometry and labels are correct.

ANSWER RELATIONS / CHECKS

Conceptual proof = principle + interpretation + evidence
Prefer sign, direction, units, and limiting-case checks

1 Electric-field word bank

1 ELECTRIC-FIELD WORD BANK

Concepts needed: field definition; test charge; vector addition; conductor equilibrium; Gauss law; inverse-square law; charge conservation

The correct words are determined by the mathematical type of each idea. Electric field is a vector because it has magnitude and direction, and it is defined using a positive test charge. "Source" identifies the charge arrangement creating the field; "test" identifies the small charge used to measure force. Coulomb's law has inverse-square dependence because spherical spreading changes the field over an area proportional to r^2 . In a conductor at equilibrium, mobile charges have rearranged until the interior field is zero. Gauss law selects enclosed charge for net flux, while an external charge can still alter local field. Symmetry can make contributions cancel or reinforce, and conservation says total charge is not created or destroyed.

1. a test; b source; c vector; d zero; e enclosed; f perpendicular; g inverse-square; h conserved; i canceled; reinforced.
2. The distractors "scalar," "tangent," "external," "nonzero," "frequency," and "wavelength" do not fit these blanks.
3. The repeated use of "test" and the two symmetry outcomes are deliberate checks against guessing from word frequency. j. positive; k. conserved. A field is a source-created property of space; force is the interaction qE when a charge is placed there. N/C is force per charge, so newtons alone would omit the test-charge dependence.

ANSWER RELATIONS / CHECKS

$$E = F/q$$

$$F = k|q_1q_2|/r^2$$

$$Q_{\text{total}}(\text{before}) = Q_{\text{total}}(\text{after})$$

2 Matching visuals to equations

2 MATCHING VISUALS TO EQUATIONS

Concepts needed: field-line geometry; Gauss flux; SHM; traveling-wave translation; potential slope; logarithmic level

Each match follows from the feature emphasized by the visual. A radial field from a point charge is the inverse-square vector field. A closed surface with a net field-line excess represents flux and enclosed charge, not merely a local field value. Oscillation about equilibrium is represented by a second-time-derivative relation with a restoring sign. A translated shape must depend on $x-vt$ for rightward motion. A potential well is interpreted through slope, so force is the negative derivative. Sound level is a logarithmic comparison with a reference, so the decibel expression is the only match that contains a logarithm and I_0 .

1. A-3: radial inverse-square field from a point charge.
2. B-4: closed-surface flux is set by enclosed charge.
3. C-6: SHM has x'' proportional to $-x$.
4. D-5: $x-vt$ translates a shape in $+x$.
5. E-1: force is negative potential slope.
6. F-2: sound level is logarithmic relative to I_0 .

ANSWER RELATIONS / CHECKS

$$E = kQ/r^2$$

$$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$$

$$x'' + \omega^2 x = 0$$

$$y = f(x - vt)$$

$$F = -dU/dr$$

$$\beta = 10 \log_{10}(I/I_0)$$

3 Explain electric-field phenomena

3 EXPLAIN ELECTRIC-FIELD PHENOMENA

Concepts needed: force direction; superposition; conductor equilibrium; flux

A negative test charge feels $F=qE$, so the negative scalar reverses the vector direction relative to E . Two nonzero fields can cancel because vectors add component-by-component; equal opposite vectors sum to zero at a symmetry point. Inside a conductor, a nonzero static field would accelerate mobile charges, contradicting equilibrium. Surface charges rearrange until the interior field cancels. The tangential component at the surface must vanish because otherwise charges would continue moving along the surface, so the remaining field is normal. Gauss law says external charge contributes as much inward and outward flux in aggregate. e. The field can exist with no test charge; the force is defined only after a charge is placed. f. Field is force divided by charge, so its unit must be N/C. as it crosses a closed surface, giving zero net contribution even though the local field is not zero.

1. a. State $F=qE$, then note $q<0$ reverses direction.
2. b. Draw equal arrows and add components; cancellation is vector addition, not absence of individual fields.
3. c. Use the contradiction with charge motion, then state normal surface field.
4. d. Distinguish local flux density from total closed-surface flux.

ANSWER RELATIONS / CHECKS

$$F=qE$$

$$E_{\text{inside metal}}=0$$

$$E_{\text{tangential}}=0$$

$$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$$

4 SHM graphs

4 SHM GRAPHS

Concepts needed: sinusoidal motion; phase; derivatives; energy conservation

For $x=A \cos(\omega t)$, the displacement begins at $+A$. Differentiate once: $v=-A\omega \sin(\omega t)$, so velocity begins at zero and becomes negative. Differentiate again: $a=-A\omega^2 \cos(\omega t)=-\omega^2 x$, so acceleration is opposite displacement and has maximum magnitude at the turning points. The spring potential is $U_s=1/2 kx^2$: an upward parabola with maximum at $\pm A$. Total energy is $1/2 kA^2$, so $K=E-U_s$ is largest at $x=0$ and zero at $\pm A$. This is why displacement, velocity, acceleration, and energy graphs are linked but not identical.

1. Draw x , v , and a with the same period; v is quarter-cycle shifted from x and a is opposite x .
2. At $x=0$: speed is maximum, $U_s=0$, and acceleration is zero. For $x<0$, acceleration and restoring force point right, toward equilibrium. Ideal SHM has an amplitude-independent period because the linear equation has a fixed ω ; amplitude changes energy, not ω .
3. At $x=\pm A$: speed is zero, U_s is maximum, and $|a|$ is maximum.
4. K is an inverted parabola relative to U_s , and $K+U_s$ is constant.

ANSWER RELATIONS / CHECKS

$$\begin{aligned}x &= A \cos(\omega t) \\v &= -A\omega \sin(\omega t) \\a &= -\omega^2 x \\U_s &= 1/2 kx^2; K = 1/2 k(A^2 - x^2)\end{aligned}$$

5 Wave-graph statements

5 WAVE-GRAPH STATEMENTS

Concepts needed: amplitude; wave number; direction; particle motion; linear medium

Statement II is correct because $k=2\pi/\lambda$, so larger k means shorter wavelength. Statement III is correct because a fixed phase point in $\cos(kx-\omega t)$ moves toward increasing x , while $\cos(kx+\omega t)$ moves toward decreasing x . Statement I is false for an ideal linear medium: amplitude controls disturbance size, while speed is set by medium properties. Statement IV is false because particles oscillate locally while the waveform transports energy at the propagation speed. The particle's transverse velocity can be larger or smaller than the wave speed. A medium point oscillates about equilibrium while the pattern carries phase and energy; a crest is a location of the pattern, not a marked piece of matter. and is a different physical quantity.

1. Select II and III only.
2. Label one wavelength between matching phase points and amplitude from equilibrium to crest.
3. Explain that changing A changes energy carried, not necessarily v in a linear medium.
4. State transverse versus longitudinal motion separately from propagation direction.

ANSWER RELATIONS / CHECKS

$$y=A \cos(kx-\omega t)$$

$$k=2\pi/\lambda$$

$$v_{\text{wave}}=\omega/k=f\lambda$$

6 Oscillation comparisons

6 OSCILLATION COMPARISONS

Concepts needed: period; mass-spring frequency; energy; resonance; damping

For a mass-spring oscillator, $T=2\pi\sqrt{m/k}$, so doubling mass increases the period by $\sqrt{2}$. At equilibrium, all ideal energy is kinetic and speed is maximum; at a turning point, speed is zero and spring energy is maximum. Total energy is $\frac{1}{2}kA^2$, so at fixed amplitude the stiffer spring has more energy. Damping removes mechanical energy, so a driven oscillator has a finite resonance peak rather than the indefinitely large ideal response. As driving frequency rises from low values, amplitude grows toward resonance and then falls above resonance; the exact peak depends on damping. Resonance is especially effective energy transfer when driving is near the system's natural frequency. Damping lowers and broadens the response; natural frequency belongs to the system, not the driver.

1. a. $T_2/T_1=\sqrt{2}$ for twice the mass.
2. b. $x=0$ gives $v_{\max}$; $x=A$ gives $v=0$ and U_s maximum.
3. c. Larger k means larger E when A is the same.
4. d. Amplitude rises, peaks near natural frequency, then falls; damping broadens and lowers the peak.

ANSWER RELATIONS / CHECKS

$$T=2\pi\sqrt{m/k}$$

$$\omega=\sqrt{k/m}$$

$$E=\frac{1}{2}kA^2$$

7 Sound and spectrum

7 SOUND AND SPECTRUM

Concepts needed: mechanical longitudinal waves; pressure; intensity; decibels; harmonics; timbre

A pure tone has one frequency component and a sinusoidal or nearly sinusoidal time trace. A periodic complex tone repeats in time but has a fundamental plus harmonics. Noise has irregular time behavior and a broad, non-discrete spectrum. In air, molecules oscillate parallel to propagation, forming compressions and rarefactions. For a sinusoidal sound, intensity is proportional to rms pressure squared, so doubling pressure makes intensity four times larger. Sound-pressure level changes by $20\log_{10}(2)=6.02$ dB. Source loudness changes amplitude, not the medium's stiffness. Amplitude is maximum displacement or pressure variation; intensity is power per area; pitch tracks frequency; loudness is perception; timbre tracks waveform and harmonic content. A spectrum displays amplitude versus frequency, density, or temperature, so it does not by itself determine sound speed.

1. Draw one spectral line for a pure tone, several harmonic lines for a complex tone, and a broad irregular spectrum for noise.
2. Mark compression as high pressure and rarefaction as low pressure.
3. Use the square relation before applying the logarithm.
4. Separate pitch, loudness, timbre, and propagation speed.

ANSWER RELATIONS / CHECKS

$$I \propto p_{\text{rms}}^2$$

$$\Delta L_p = 20 \log_{10}(p_2/p_1)$$

v_{sound} depends on medium properties

8 Waves at boundaries

8 WAVES AT BOUNDARIES

Concepts needed: frequency continuity; wavelength; Huygens principle; refraction; normal angle

At a stationary boundary, the source fixes frequency, so frequency stays continuous across the interface. The new medium sets a new speed, and $\lambda=v/f$ therefore changes. Huygens' principle says each point on a wavefront emits secondary wavelets; when one side of a wavefront slows first, the wavefront pivots and the direction changes. Angles are measured from the normal because the normal is the direction in which the boundary crossing and phase matching are compared. If $n_2 > n_1$, then $v_2 < c/n_1$ is smaller, so the ray bends toward the normal. Transverse and longitudinal describe particle motion relative to propagation; sound in air is longitudinal and requires a material medium. Electromagnetic waves do not require matter. At a stationary boundary, frequency remains source-set while speed and wavelength can change.

1. a. In a slower medium: frequency unchanged, speed lower, wavelength shorter, ray toward normal.
2. b. Draw successive wavelets and their envelope; the unequal advance rotates the envelope.
3. c. Use $n=c/v$ and the phase-matching sine relation.
4. d. State that mechanical waves require a medium; electromagnetic waves do not.

ANSWER RELATIONS / CHECKS

$$v=f\lambda$$

$$n=c/v$$

$$n_1 \sin\theta_1=n_2 \sin\theta_2$$

9 Symmetry reasoning

9 SYMMETRY REASONING

Concepts needed: inverse-square scaling; superposition; midpoint symmetry

For one point charge at r , $E=k|Q|/r^2$. At $2r$, the field is one quarter as large. For two equal like charges at the midpoint, the two field vectors point in opposite directions and cancel, so the net field is zero. For equal opposite charges, the fields at the midpoint point in the same direction—from the positive charge toward the negative charge—so they reinforce. The comparison illustrates why vector direction must be drawn before magnitudes are combined. A zero result can arise from cancellation even though each individual contribution is nonzero. Off the midpoint, the exact symmetry cancellation is lost, so the geometry and component calculation are needed; inverse-square scaling alone is insufficient.

1. (ii) is one quarter of (i).
2. (iii) is zero by opposite-vector cancellation.
3. (iv) is nonzero and points from + toward -.
4. Do not rank (iv) against (i) numerically without a separation specification; state the direction and scaling instead.

ANSWER RELATIONS / CHECKS

$$E_{\text{point}}=k|Q|/r^2$$

$$E_{\text{net}}=\Sigma E_i$$

$$E(r=2r)=E(r)/4$$

10 Integrated concept check

10 INTEGRATED CONCEPT CHECK

Concepts needed: scalar potential; vector field; work; phase; pitch; spectrum

Potential is a scalar because it has magnitude but no direction; electric field is a vector. Work along an equipotential is zero because $dV=0$, and E points toward decreasing potential. A wave's phase changes with both position and time, through kx and ωt . In sound, pitch is primarily frequency. Timbre depends on waveform and harmonic spectrum. f. Capacitance describes charge per unit potential difference. g. A dipole can have nonzero torque even when its uniform-field net force is zero., which is why two instruments can have the same fundamental pitch but sound different.

1. a. scalar; vector. b. zero. c. lower. d. position; time. e. frequency; spectrum.
2. Explain one: $E = -\text{grad } V$ means the field follows the steepest decrease in potential.
3. For phase, compare $kx - \omega t$ at two points or times.

ANSWER RELATIONS / CHECKS

$$\begin{aligned} dV &= -\int E \cdot dl \\ E &= -\text{grad } V \\ \text{phase} &= kx - \omega t \end{aligned}$$

11 Comprehensive explanation prompt

11 COMPREHENSIVE EXPLANATION PROMPT

Concepts needed: synthesis; model selection; diagrams; physical interpretation

Answers vary, but a strong response must connect representations. For A, begin with the point-charge field and field-line rules, then define flux and explain why a Gaussian surface is useful only with symmetry, and finally connect conductor shielding to the zero interior field. For B, distinguish local oscillation from energy transport: SHM describes one oscillator, a traveling wave translates a disturbance, and sound is a mechanical longitudinal wave whose pressure variations carry energy. For C, decode each part of $y=A \cos(kx-\omega t+\phi)$: A is amplitude, k sets wavelength, ω sets frequency, ϕ sets phase, and the sign determines direction. A correct diagram must reinforce, not replace, the explanation.

1. A should include $E=F/q$, $\Phi=\int E \cdot dA$, Gauss law, and $E=0$ in metal.
2. B should include $x(t)$, $y(x,t)$, compression/rarefaction, and $v=f\lambda$.
3. C should label A, λ , phase, and a right-moving crest.
4. Award full credit when the response states assumptions and checks direction or limiting behavior. Strong responses should also connect the requested limiting case to a concrete prediction rather than merely naming it.

ANSWER RELATIONS / CHECKS

$$\Phi = \int E \cdot dA = Q_{\text{enc}}/\epsilon_0$$

$$x = A \cos(\omega t + \phi)$$

$$y = A \cos(kx - \omega t + \phi)$$

$$v = f\lambda$$